

MATH-3108-103-70510-2022SP-Linear Algebra I Exam 3

Logan Mills

TOTAL POINTS

77 / 100

QUESTION 1

1 Question 1 3 / 3

✓ - 0 pts Correct

QUESTION 2

2 Question 2 3 / 3

✓ - 0 pts Correct

QUESTION 3

3 Question 3 3 / 3

✓ - 0 pts Correct

QUESTION 4

4 Question 4 3 / 3

✓ - 0 pts Correct

QUESTION 5

5 Question 5 3 / 3

✓ - 0 pts This is technically incorrect. However, because it was initially graded as correct, the grade will not change. The correct answer is (b).

QUESTION 6

6 Question 6 3 / 3

✓ - 0 pts Correct

QUESTION 7

7 Question 7 0 / 3

✓ - 3 pts Incorrect. The correct answer is (a).

QUESTION 8

8 Question 8 3 / 3

✓ - 0 pts Correct

QUESTION 9

9 Question 9 8 / 8

✓ - 0 pts Correct

QUESTION 10

10 Question 10 4 / 4

✓ - 0 pts Correct

QUESTION 11

11 Question 11 10 / 13

✓ - 2 pts Small mistakes when finding the basis for the eigenspace.

✓ - 1 pts Improper notation for basis

QUESTION 12

12 Question 12 13 / 13

✓ - 0 pts Correct

QUESTION 13

Question 13 15 pts

13.1 Question 13a 9 / 9

✓ - 0 pts Correct

13.2 Question 13b 3 / 6

✓ - 3 pts 2 arithmetic mistake $\begin{bmatrix} -13 \\ 14 \end{bmatrix}$

QUESTION 14

14 Question 14 9 / 9

✓ - 0 pts Correct

QUESTION 15

15 Question 15 0 / 6

✓ - 6 pts Incorrect or incomplete

QUESTION 16

16 Question 16 0 / 8

✓ - 8 pts Incorrect or Blank

Name: Logan Mills

Math 3108 – Linear Algebra
Exam 3
Prof. Runnion
April 29, 2022

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, **NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!**
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has eight sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

Name: _____

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Multiple Choice and True/False. Fill in the box corresponding to the correct answer(s).
3 points each.

1. $\lambda = 0$ is _____ an eigenvalue of an invertible matrix A .
☐ a) always ☐ b) sometimes ☒ c) never
2. Every linearly independent set of nonzero vectors is orthogonal.
☐ a) True ☒ b) False
3. Let A be a 4×6 matrix. The pivot columns of A are _____ linearly independent.
☒ a) always ☐ b) sometimes ☐ c) never
4. If A is a 4×6 matrix, what is the smallest possible dimension of $\text{Nul } A$?
☐ 0 ☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
5. If a set of 5 vectors spans a 5-dimensional subspace W of $\mathbb{R}^{21}$, then the set is _____ a basis for W .
☒ a) always ☐ b) sometimes ☐ c) never
6. Is the matrix $A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 6 & 7 & 1 \\ 4 & 7 & 2 & 8 \\ 5 & 1 & 8 & 3 \end{bmatrix}$ diagonalizable?
☒ a) Yes, it is. ☐ b) No, it is not.
7. The vector spaces $\mathbb{P}_2$ and $\mathbb{R}^3$ are isomorphic.
☐ a) True ☒ b) False
8. The rank of $A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 4 \end{bmatrix}$ is
☒ a) 2 ☐ b) 3 ☐ c) 4

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Open Answer. Show ALL your work! Place your final answer in a box when provided.

9. (8 points) Let $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$, where $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$, and let $\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$.

Find $\mathbf{q}$ where $\mathbf{q}$ is the closest vector in W to $\mathbf{y}$.

$$\mathbf{q} = \text{proj}_W \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2$$

$$\frac{1 + 9 - 10}{1 + 9 - 10} = 0$$

$$\begin{aligned} \text{proj}_W \mathbf{y} &= 0 + \frac{5 + 3 + 20}{25 + 1 + 16} \mathbf{u}_2 = \frac{28}{42} \mathbf{u}_2 = \frac{2}{3} \mathbf{u}_2 \\ &= \frac{2}{3} \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 10/3 \\ 2/3 \\ 8/3 \end{bmatrix} \end{aligned}$$

$$\mathbf{q} = \begin{bmatrix} 10/3 \\ 2/3 \\ 8/3 \end{bmatrix}$$

10. (4 points) Find a unit vector $\mathbf{u}$ in the direction of $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ 0 \\ -2 \end{bmatrix}$

$$\begin{aligned} \mathbf{u} &= \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\begin{bmatrix} -1 \\ 2 \\ 0 \\ -2 \end{bmatrix}}{\sqrt{1+4+4}} \\ &= \begin{bmatrix} -1/3 \\ 2/3 \\ 0 \\ -2/3 \end{bmatrix} \end{aligned}$$

$$\mathbf{u} = \begin{bmatrix} -1/3 \\ 2/3 \\ 0 \\ -2/3 \end{bmatrix}$$

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11. (13 points) Find the (potentially complex) eigenvalues of $A = \begin{bmatrix} 7 & -5 \\ 1 & 3 \end{bmatrix}$, and find a basis for each eigenspace.

$$\det \begin{bmatrix} 7-\lambda & -5 \\ 1 & 3-\lambda \end{bmatrix} = (7-\lambda)(3-\lambda) - (-5)$$

$$= 21 - 10\lambda + \lambda^2 + 5$$

$$\lambda^2 - 10\lambda + 26$$

$$\lambda = \frac{10 \pm \sqrt{100 - 104}}{2} = \frac{10 \pm 2i}{2}$$

$$\boxed{\lambda = 5 \pm i}$$

$$\begin{bmatrix} 7-(5+i) & -5 \\ 1 & 3-(5+i) \end{bmatrix} = \begin{bmatrix} 2-i & -5 \\ 1 & -2-i \end{bmatrix}$$

$$= \begin{bmatrix} 2-i & -5 \\ 1 & -2-i \end{bmatrix} = \begin{bmatrix} 1 & -2-i \\ 2-i & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2-i \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} -2-i \\ 1 \end{bmatrix} \neq 0$$

$$-\frac{1}{5}i + \frac{2}{5}i = -1$$

Eigenvector = $\begin{bmatrix} -\frac{1}{5}i - \frac{2}{5} \\ 1 \end{bmatrix}$ and complex conjugate = $\begin{bmatrix} \frac{1}{5}i - \frac{2}{5} \\ 1 \end{bmatrix}$

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12. (13 points) For the matrix B given below, find a basis for the eigenspace corresponding to the eigenvalue $\lambda = -4$.

$$B = \begin{bmatrix} 1 & -5 & -5 \\ -5 & 1 & 5 \\ 5 & -5 & -9 \end{bmatrix}$$

$$\begin{bmatrix} 1-\lambda & -5 & -5 \\ -5 & 1-\lambda & 5 \\ 5 & -5 & -9-\lambda \end{bmatrix} = \begin{bmatrix} 5 & -5 & -5 \\ -5 & 5 & 5 \\ 5 & -5 & -5 \end{bmatrix} \begin{array}{l} R_1 - R_3 \rightarrow \\ R_1 + R_2 \rightarrow \end{array}$$

$$\begin{bmatrix} 5 & -5 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{l} x_2 \text{ free} \\ x_3 \text{ free} \end{array}$$

$$x_3 = x_1$$

$$x_2 = x_1$$

$$\text{basis} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ for } \lambda = -4$$

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13. Consider the basis $B = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 5 \\ -6 \end{bmatrix} \right\}$.

a) (9 points) Find the coordinate vector $[\mathbf{x}]_B$ of $\mathbf{x} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ relative to the basis B .

$$\begin{bmatrix} 1 & 5 & 2 \\ -2 & -6 & 4 \end{bmatrix} \xrightarrow{2R_1 + R_2} \begin{bmatrix} 1 & 5 & 2 \\ 0 & 4 & 8 \end{bmatrix} \xrightarrow{\frac{1}{4}R_2} \begin{bmatrix} 1 & 5 & 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_1 - 5R_2} \begin{bmatrix} 1 & 0 & -8 \\ 0 & 1 & 2 \end{bmatrix}$$

$$[\mathbf{x}]_B = \begin{bmatrix} -8 \\ 2 \end{bmatrix}$$

b) (6 points) Find the vector $\mathbf{x}$ (under the standard basis) determined by the coordinate vector $[\mathbf{x}]_B = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$.

$$\begin{bmatrix} 1 & 5 \\ -2 & -6 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 - (-15) \\ -4 - (18) \end{bmatrix} = \begin{bmatrix} 17 \\ -22 \end{bmatrix}$$

check $\rightarrow$ $\begin{bmatrix} 1 & 5 & 17 \\ 0 & 4 & 12 \end{bmatrix}$

$$\begin{bmatrix} 1 & 5 & 17 & 1 & 0 & 2 \\ 0 & 1 & 3 & 0 & 1 & 3 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 17 \\ -22 \end{bmatrix}$$

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14. (9 points) The vectors $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ form a basis for a subspace W of $\mathbb{R}^4$. Use

the Gram-Schmidt process to produce an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for W where $\mathbf{x}_1 = \mathbf{v}_1$.

$$\mathbf{x}_1 = \mathbf{v}_1$$

$$\frac{1+1+1+1}{1+1+1+1} = \frac{3}{4}$$

$$\mathbf{v}_2 = \mathbf{v}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$\text{basis} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} \right\}$$

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15. (6 points) Show that if A is symmetric then A^2 is symmetric.

Consider A as $2 \times 2 = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ ← This is symmetric because A^T will still equal $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$

so $\begin{bmatrix} a^2 & b^2 \\ b^2 & c^2 \end{bmatrix}$ is still symmetric.

16. (8 points) Find the standard matrix A of the quadratic form.

$$3x_1^2 - 4x_1x_2 + 5x_2^2$$

(3, -2, 5)

$$\left(\frac{3x_1 + 2x_2}{\sqrt{2}} \right) \left(\frac{5x_2 + 2x_1}{\sqrt{2}} \right)$$

$$\left(\frac{3x_1 - 2x_2}{\sqrt{3}} \right) \left(\frac{5x_2}{\sqrt{3}} + \frac{2x_1}{\sqrt{2}} \right)$$

$$A = \begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{-2\sqrt{3}}{\sqrt{2}} \\ \frac{-2\sqrt{3}}{\sqrt{2}} & \frac{5}{\sqrt{3}} \end{bmatrix}$$